Introduction to Biophysics II: Quantum physics & biology

Lecture XX: Circular dichroism

Circular polarization
Circular dichroism: units
Molecular symmetry, handedness, biology
Absorption strength difference of right and left circularly polarized light
- Interaction with magnetic and electric transition dipoles
- Magnetic transition dipole
- Excitonic circular dichroism

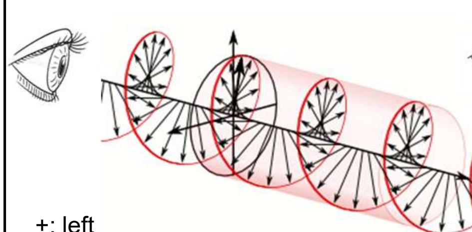
Making circularly polarized light
CD spectrometer design

Reading:
Modern optical spectroscopy, chapter 10 (the 'green' book)
Excitons, van Amerongen et al.

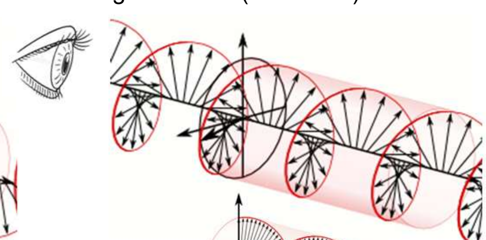
1

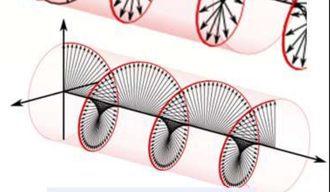
Circularly polarized light

Left-handed (counter-clockwise)



Right-handed (clockwise)

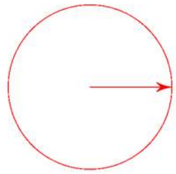




+: left
-: right

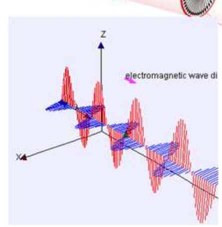
$$\vec{E}^{\pm}(t,z) = E_0\left[\cos(\omega t - kz)\hat{x} \pm \sin(\omega t - kz)\hat{y}\right]$$

$$\omega = \frac{2\pi c}{\lambda} \qquad k = \frac{2\pi}{\lambda}$$



Linear polarization:

$$\vec{E}(t,z) = E_0\cos(\omega t - kz)\hat{x}$$
$$\vec{B}(t,z) = B_0\cos(\omega t - kz)\hat{y}$$
$$B_0 = E_0 / c$$



2

Circularly polarized light

Linear: $\vec{E}_{lin}(t,z) = E_0 \cos(\omega t - kz)\hat{x}$ $\quad \omega = \frac{2\pi c}{\lambda} \quad k = \frac{2\pi}{\lambda}$

+: Left
-: Right $\quad \vec{E}^{\pm}(t,z) = E_0\left[\cos(\omega t - kz)\hat{x} \pm \sin(\omega t - kz)\hat{y}\right]$

In this representation, the energy density of circularly polarized light is twice that of linearly polarized light:

$$u_{lin} = \tfrac{1}{2}\epsilon_0 E_0^2 \qquad u_{circ} = \epsilon_0 E_0^2$$

Often, an alternative equation is used where the energy density of circularly polarized light is the same as that of linearly polarized light:

$$\vec{E}^{\pm}(t,z) = \tfrac{1}{\sqrt{2}} E_0\left[\cos(\omega t - kz)\hat{x} \pm \sin(\omega t - kz)\hat{y}\right] \Rightarrow u_{circ} = \frac{1}{2}\varepsilon_0 E_0^2$$

Alternatively, using complex space:

$$\begin{cases} \tilde{E}^{\pm} = E_0(\hat{x} \pm i\hat{y})e^{-i\omega t} \\ \tilde{B}^{\pm} = \frac{1}{c}\hat{z} \times \vec{E}^{\pm} = \mp \frac{i}{c}\vec{E}^{\pm} \end{cases}$$

OR:

$$\begin{cases} \tilde{E}^{\pm} = \tfrac{1}{\sqrt{2}} E_0(\hat{x} \pm i\hat{y})e^{-i\omega t} \\ \tilde{B}^{\pm} = \frac{1}{c}\hat{z} \times \vec{E}^{\pm} = \mp \frac{i}{c}\vec{E}^{\pm} \end{cases}$$

Note: linear polarization: $\vec{E}_{lin} = \frac{1}{\sqrt{2}}\left(\tilde{E}^{+} + \tilde{E}^{-}\right) = E_0 \hat{x} e^{i(kz - \omega t)}$

3

Circular dichroism (CD)

Circular Dichroism: Absorption of circularly polarized light depends on the handedness of light, L (left) or R (right)

$$A_L \equiv A^{+} = A + \Delta A$$
$$A_R \equiv A^{-} = A - \Delta A$$

A: Conventional absorption
ΔA: Change due to circular polarization

CD in optical density units: $\boxed{\Delta A_{CD} = A_L - A_R = 2\Delta A}$ It depends on the wavelength

Molar extinction CD: $\Delta\varepsilon_{CD}(\lambda) = \varepsilon_L(\lambda) - \varepsilon_R(\lambda) = \Delta A_{CD}/(cl)$ *c – molar concentration*
l – optical thickness

CD measured as ellipticity in degrees:
Recall linear polarization: $\vec{E}_{lin} = \frac{1}{\sqrt{2}}\left(\tilde{E}^{+} + \tilde{E}^{-}\right) = E_0 \hat{x} e^{i(kz - \omega t)}$

- Linear polarization is an equal superposition of L and R-polarized light
- The sample absorbs one more than the other one
  → After the sample polarization is elliptical

$$\begin{cases} E_x(z,t) = E_0 \cos\theta \cos(kz - \omega t) \\ E_y(z,t) = -E_0 \sin\theta \cos(kz - \omega t) \end{cases}$$

Ellipticity:

Ellipticity θ:

For small θ (and ΔA)

$$\boxed{\tan\theta = \frac{E_R - E_L}{E_R + E_L} \approx \theta[rad] \approx \ln(10)\frac{\Delta A_{CD}}{4}}$$

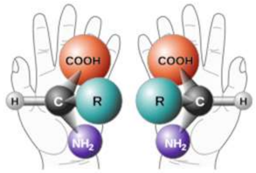
Unit conversion

$$\theta[\mathrm{mdeg}] \approx 33 \times 10^3 \Delta A_{CD}$$

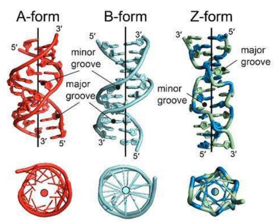
4

The origin of CD (chirality)

Two enantiomers of a generic amino acid that is chiral (handedness)



DNA



Alpha helix

Most amino acids have two enantiomers:
Left-handed: denoted S (Sinistral) or L (Left)
Right-handed: denoted D (Dextral) or R (Right)

Natural amino acids are mostly of L-type; D-type is rare.

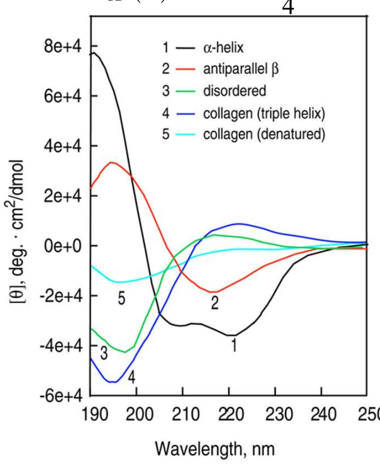
Alpha helices in proteins, as well as DNA, are almost always right-handed

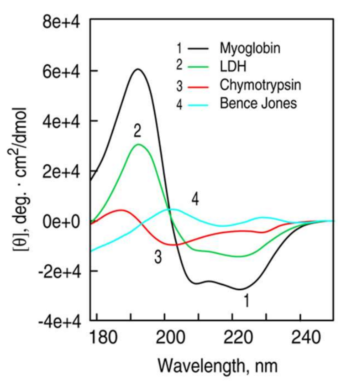
Circular Dichroism: Absorption of circularly polarized light depends on the handedness of light and molecules

Examples of Circular Dichroism Spectra

The difference in absorption of left and right circularly polarized light

$$\Delta\varepsilon_{CD}(\lambda) = \ln(10)\frac{\Delta A_{CD}}{4} / (cl) \qquad \boxed{\Delta A_{CD} = A_L - A_R}$$

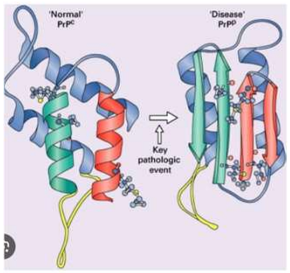




CD spectra of proteins with defined secondary structure (left) and CD spectra of representative proteins with varying conformations (right). Greenfield, N. *Nature Protocols*, p. 2876, v. 1, 2007.

https://www.bif.mit.edu/new-page-1-1

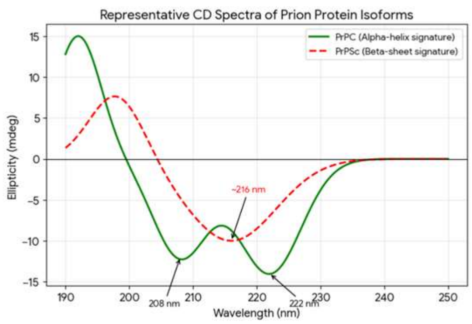
Example: using CD signals to study proteins

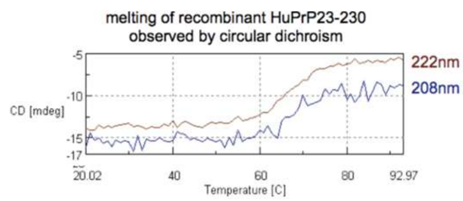


PrP: A major prion protein in the nervous system

It is crucial for neuron health and has been linked to synaptic formation and function.

Prion diseases occur when \(\text{PrP}^{\text{C}}\) transforms into an abnormal, pathogenic isoform (\(\text{PrP}^{\text{Sc}}\)), forming infectious amyloid fibrils that cause neurodegeneration, such as Creutzfeldt-Jakob disease (CJD) in humans and scrapie in animals.





7

Accounting for the magnetic dipole moment

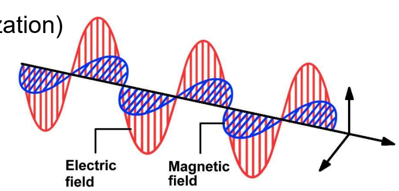
The intrinsic CD of a molecule is the result of the interaction of a molecule with both the electric and magnetic fields of a light electromagnetic wave

Absorption of light, earlier lecture: interaction with light $\tilde{\mathrm{H}}'(t) = -\boldsymbol{E}(t)\cdot\tilde{\mu}$

Light also has a magnetic field component

$$\begin{cases} \vec{E} = E_0 e^{i(kz-\omega t)}\hat{x} \\ \vec{B} = B_0 e^{i(kz-\omega t)}\hat{y} = \dfrac{1}{c}\hat{z}\times\vec{E} \end{cases}$$

(Linear polarization)



Magnetic field is weak: $B_0 = \dfrac{E_0}{c}$

A molecule may have a magnetic moment (orbital motion of charges, and spins)
Include interaction with the magnetic component:

$$\tilde{\mathrm{H}}'(t) = -\boldsymbol{E}(t)\cdot\tilde{\mu} - \boldsymbol{B}(t)\cdot\tilde{\mathrm{m}}$$

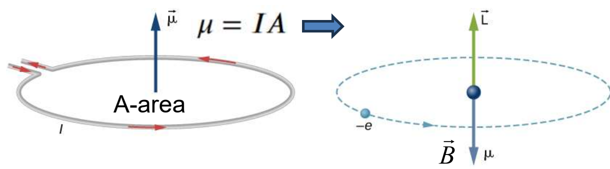
Magnetic moment operator

8

Magnetic (dipole) moment

$$\tilde{H}'(t) = -\boldsymbol{E}(t)\cdot\tilde{\mu} - \boldsymbol{B}(t)\cdot\tilde{m}$$

Magnetic field of a molecule due to orbital motion of electrons (Bohr model):



(a) Current-carrying loop

(b) Hydrogen atom

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

$$A = \pi r^2$$

$$\mu = IA = \frac{ev}{2\pi r}\pi r^2 = \frac{evr}{2}$$

SI: $\vec{\mu} = \frac{-e}{2}\left(\vec{v}\times\vec{r}\right) = \frac{-e}{2m_e}\left(\vec{p}\times\vec{r}\right)$

CGS: $\vec{\mu} = \frac{-e}{2m_e c}\left(\vec{p}\times\vec{r}\right)$

Magnetic moment: μ
determines the torque an object experiences in an external magnetic field

Magnetic moment operator:

Electron *g-factor* = 2.00232

Electron spin: negligible contribution → ignore

$$\tilde{m} = \frac{e}{2m_e c}(\boldsymbol{r}\times\tilde{p} + g_e\boldsymbol{S})$$

Linear momentum operator: $\tilde{p} = -i\hbar\tilde{\nabla}$

9

Magnetic moment-driven transitions

$$\tilde{H}(t) = -\vec{B}(t)\cdot\tilde{m} \qquad \tilde{m} = \frac{e}{2m_e c}\vec{r}\times\tilde{p}$$

Transition between states *a* and *b*: need matrix elements:

$$\tilde{H}'_{ba} \equiv \left\langle \psi_b \middle| \tilde{H}' \middle| \psi_a \right\rangle$$

$$\tilde{H}'_{ba} = -\vec{B}\cdot\left\langle \psi_b \middle| \tilde{m} \middle| \psi_a \right\rangle$$

This is analogous to transition driven by electric field $\tilde{H}'_{ba} = -\vec{E}\cdot\left\langle \psi_b \middle| \tilde{\mu} \middle| \psi_a \right\rangle$

We showed earlier (Lec. 3-4), that the transition probability is $\propto \left|\vec{E}_0\cdot\vec{\mu}_{ba}\right|^2$

$$\vec{\mu}_{ba} \equiv \left\langle \psi_b \middle| \tilde{\mu} \middle| \psi_a \right\rangle$$

Similarly, the transition probability due to the magnetic field is $\propto \left|\vec{B}\cdot\vec{m}_{ba}\right|^2$

$$\vec{m}_{ba} = \left\langle \psi_b \middle| \tilde{m} \middle| \psi_a \right\rangle$$

Note: $\left|\vec{m}_{ba}\right| << \left|\vec{\mu}_{ba}\right|$ *for chlorophylls and dyes in general*

The absorption strength driven by the magnetic component of light is about 2-4 orders of magnitude weaker than that driven by the electric component

10

Magnetic transition dipole moment

Consider the transition from HOMO (ψ_h) to LUMO (ψ_l)

Ground state $\Psi_a = \psi_h(1)\psi_h(2)$

Excited state $\Psi_b = 2^{-1/2}\psi_h(1)\psi_l(2) + 2^{-1/2}\psi_h(2)\psi_l(1)$

Since electrons are indistinguishable, use the Slater-type function for the excited state

Find: $\vec{m}_{ba} \equiv \langle \psi_b | \tilde{m} | \psi_a \rangle$ $\qquad \tilde{m} = \frac{e}{2m_e c}\vec{r} \times \tilde{p}$

$$\vec{m}_{ba} = \frac{1}{\sqrt{2}}\left\langle \psi_h(1)\psi_l(2) + \psi_h(2)\psi_l(1) \middle| \frac{e}{2m_e c}\left(\vec{r} \times \{-i\hbar\tilde{\nabla}\}\right) \middle| \psi_h(1)\psi_h(2) \right\rangle$$

$$\vec{m}_{ba} = -i\hbar\frac{1}{\sqrt{2}}\frac{e}{2m_e c}\left[\langle \psi_h(1)\psi_l(2)|\vec{r}\times\tilde{\nabla}|\psi_h(1)\psi_h(2)\rangle + \langle \psi_h(2)\psi_l(1)|\vec{r}\times\tilde{\nabla}|\psi_h(1)\psi_h(2)\rangle\right]$$

Acts only on transition $h \rightarrow l$

$$\vec{m}_{ba} = -i\hbar\frac{1}{\sqrt{2}}\frac{e}{2m_e c}\left[\langle \psi_h(1)|\psi_h(1)\rangle\langle \psi_l(2)|\vec{r}\times\tilde{\nabla}|\psi_h(2)\rangle + \langle \psi_h(2)|\psi_h(2)\rangle\langle \psi_l(1)|\vec{r}\times\tilde{\nabla}|\psi_h(1)\rangle\right]$$

$$\vec{m}_{ba} = -i\hbar\frac{1}{\sqrt{2}}\frac{e}{2m_e c}\left[\langle \psi_l(2)|\vec{r}\times\tilde{\nabla}|\psi_h(2)\rangle + \langle \psi_l(1)|\vec{r}\times\tilde{\nabla}|\psi_h(1)\rangle\right]$$

Terms are equal.
The only difference is the integration variable notation

$$\boxed{\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\langle \psi_l|\vec{r}\times\tilde{\nabla}|\psi_h\rangle}$$

Magnetic dipole moment is imaginary!

11

Magnetic transition dipole moment

CAN SKIP

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\langle \psi_l|\vec{r}\times\tilde{\nabla}|\psi_h\rangle$$

Consider a molecule with an orbital constructed as a linear combination of *2p* orbitals

HOMO: $\psi_h = \sum_t C_t^h p_t$

LUMO: $\psi_l = \sum_s C_s^l p_s$

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\left\langle \sum_s C_s^l p_s \middle| \vec{r}\times\tilde{\nabla} \middle| \sum_t C_t^h p_t \right\rangle$$

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\langle p_s|\vec{r}\times\tilde{\nabla}|p_t\rangle$$

12

Magnetic transition dipole moment

CAN SKIP

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left\langle p_s\left|\vec{r}\times\tilde{\nabla}\right|p_t\right\rangle$$

We will now split the vector ***r*** into the sum of vector $\boldsymbol{R_t}$ to the center of atom *t* and the position of electron $\boldsymbol{r_t}$ with respect to that: $\vec{r}\rightarrow\vec{R}_t+\vec{r}_t$

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left\langle p_s\left|\left(\vec{r}_t+\vec{R}_t\right)\times\tilde{\nabla}\right|p_t\right\rangle$$

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left[\left\langle p_s\left|\vec{r}_t\times\tilde{\nabla}\right|p_t\right\rangle+\left\langle p_s\left|\vec{R}_t\times\tilde{\nabla}\right|p_t\right\rangle\right]$$

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left[\left\langle p_s\left|\vec{r}_t\times\tilde{\nabla}\right|p_t\right\rangle+\vec{R}_t\times\left\langle p_s\left|\tilde{\nabla}\right|p_t\right\rangle\right]$$

13

Magnetic transition dipole moment

CAN SKIP

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left[\left\langle p_s\left|\vec{r}_t\times\tilde{\nabla}\right|p_t\right\rangle+\vec{R}_t\times\left\langle p_s\left|\tilde{\nabla}\right|p_t\right\rangle\right]$$

positions of electrons with respect to the atom

The overlap between the 2p orbitals of the same atom is the main contributor

Relative positions of atoms
For orbitals on the same atom, R_t=0
For orbitals on different atoms, the overlap integral is small
→ Ignore this term

The first term cross product:

$$\left\langle p_s\left|\mathbf{r}_t\times\tilde{\nabla}\right|p_t\right\rangle = \boxed{\left\langle p_s\left|y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right|p_t\right\rangle\hat{x}} - \left\langle p_s\left|x\frac{\partial}{\partial z}-z\frac{\partial}{\partial x}\right|p_t\right\rangle\hat{y} + \left\langle p_s\left|x\frac{\partial}{\partial y}-y\frac{\partial}{\partial x}\right|p_t\right\rangle\hat{z}$$

Consider x-component

$$\left\langle p_y\left|y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right|p_z\right\rangle = \left\langle p_y\left|y\frac{\partial p_z}{\partial z}\right.\right\rangle - \left\langle p_y\left|z\frac{\partial p_z}{\partial y}\right.\right\rangle$$

14

Magnetic transition dipole moment

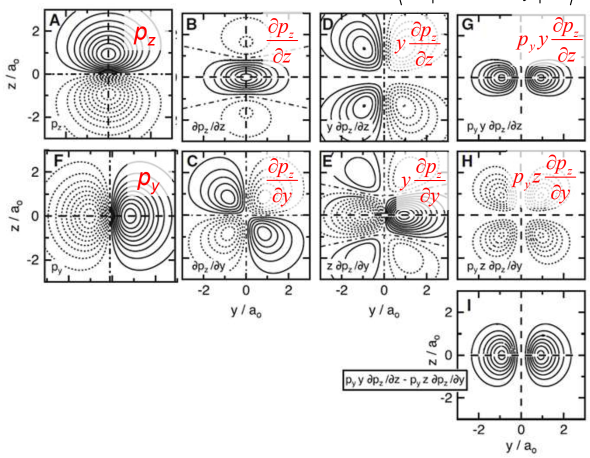
CAN SKIP

Let's investigate the first term (x-component) when

p_s : *2p_y orbital*

p_t : *2p_z orbital*

$$\left\langle p_y \middle| y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \middle| p_z \right\rangle = \left\langle p_y \middle| y\frac{\partial p_z}{\partial z} \right\rangle - \left\langle p_y \middle| z\frac{\partial p_z}{\partial y} \right\rangle$$



Opposite signs

See the comprehensive discussion in the "green" book

Combine constructively

Spatial distribution is the same as for 2p_y:

$$\langle 2p_{y(t)} | 2p_{y(t)} \rangle \propto 1$$

X-component is non-zero

Can show that for these orbitals, the *y* and *z* components are zero

→ **Magnetic moment is oriented along the x-axis for these orbitals**

15

Magnetic transition dipole moment

CAN SKIP

$$\vec{m}_{ba} = -i\frac{\sqrt{2}\hbar e}{2m_e c}\sum_s\sum_t C_s^{l*}C_t^h\left[\left\langle p_s \middle| \vec{r}_t \times \tilde{\nabla} \middle| p_t \right\rangle + \vec{R}_t \times \left\langle p_s \middle| \tilde{\nabla} \middle| p_t \right\rangle\right]$$

$$\left\langle p_s \middle| \mathbf{r}_t \times \tilde{\nabla} \middle| p_t \right\rangle = \left\langle p_s \middle| y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \middle| p_t \right\rangle \hat{x} - \left\langle p_s \middle| x\frac{\partial}{\partial z} - z\frac{\partial}{\partial x} \middle| p_t \right\rangle \hat{y} + \left\langle p_s \middle| x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \middle| p_t \right\rangle \hat{z}$$

For p_y and p_z orbitals on the same atom, $\mathbf{m}_{ba}$ is oriented along the x-axis.

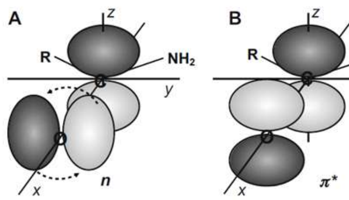
If the p_y and p_z are on different atoms, the orientation of $\mathbf{m}_{ba}$ is the same, but with a much lower amplitude due to poor overlap of the wavefunctions

16

Magnetic transition moment: example

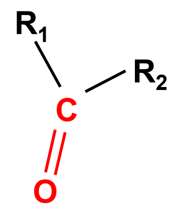
CAN SKIP

Excitation of a **carbonyl group** involves $2p_y$ and $2p_z$ orbitals:
Transition *n*-π^*, from the nonbonding orbital of the oxygen atom to π^* molecular orbital constructed of oxygen and carbon 2p orbitals.



Ground state

Excited state



$$p_{y(O)} \longrightarrow 2^{-1/2}(p_{z(C)} - p_{z(O)})$$

=0, due to symmetry

Electric transition moment:

$$\vec{\mu}_{ba} = 2^{-1/2}\left\langle \left(p_{z(C)} - p_{z(O)}\right)|\tilde{\mu}|p_{y(O)}\right\rangle = 2^{-1/2}\left\langle p_{z(C)}|\tilde{\mu}|p_{y(O)}\right\rangle - 2^{-1/2}\left\langle -p_{z(O)}|\tilde{\mu}|p_{y(O)}\right\rangle$$

Magnetic transition moment:

$$\vec{m}_{ba} = 2^{-1/2}\left\langle \left(p_{z(C)} - p_{z(O)}\right)|\tilde{m}|p_{y(O)}\right\rangle$$ Is not zero (as shown in prev. slide)

The extinction coefficient is small, <100 (for electric transitions in dyes, it is >10,000)

17

Magnetic transition moment: properties

The magnetic transition moment is weak
In most molecules, absorption due to the magnetic transition dipole moment is 2-4 orders of magnitude weaker than absorption due to the electric transition dipole moment.

Right-hand rule
The direction of **m** is determined by the "rotation" of charge density during the transition

Planar molecules have m points out of the plane
For flat (planar) molecules (such as chlorophyll), the magnetic transition dipole moment typically involves "circular" movement of electrons within the plane of the molecule (such as $\pi\rightarrow\pi^*$ transitions).

Symmetry:
In some planar molecules, transitions that are electrically forbidden **(**μ=0) can be magnetically allowed, if the transition involves a change in orbital angular momentum that mimics a current loop

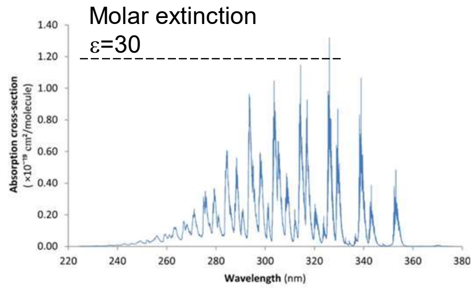
18

Magnetic vs electric transition dipole absorption

Magnetic transition

Formaldehyde ($\boldsymbol{CH_2O}$):

Its n→π* transition, the transition is strictly electric-dipole forbidden by symmetry. It becomes observable primarily through magnetic dipole transitions

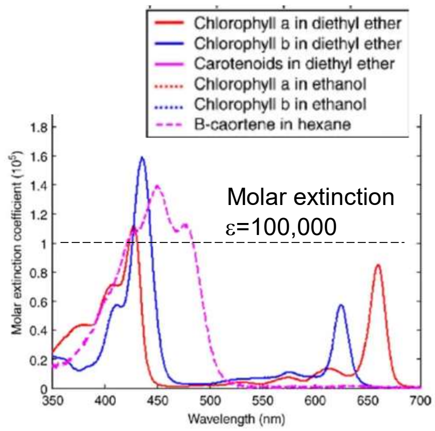


Formaldehyde absorption spectrum in the UV range, re-plotted from the data of Meller et al [30]. Peaks are not fully resolved at the resolution of this spectrum (0.01 nm).

https://www.researchgate.net/figure/Formaldehyde-absorption-spectrum-in-the-UV-range-re-plotted-from-the-data-of-Meller-et_fig1_287158085

Electric transition

Chlorophylls & carotenoids



https://www.researchgate.net/figure/upper-Differences-in-absorption-spectra-of-chlorophyll-a-chlorophyll-b-and-b-carotene_fig1_40799750

19

Interaction with circularly polarized light

*Given the right combination of μ and **m**, absorption of R and L polarization is different → circular dichroism*

Recall lecture 3:
Consider a two-state wavefunction: $\Psi = c_a(t)\Psi_a + c_b(t)\Psi_b$
Where *a* and *b* are ground and excited states, respectively.

Hamiltonian: $\tilde{H} = \tilde{H}_0 + \tilde{H}'$

At time 0 the system is in the ground state Ψ_a, i.e., c_a=1, c_b=0
Solve TDSE $\Psi_a \rightarrow \Psi_b$ and obtain $c_b(t)$ – the $|c_b(t)|^2$ is the probability of transition:

(see lecture 3, the 1st order solution):

$$c_b^{(1)}(T) = -\frac{i}{\hbar}\int_0^T \left\langle \Psi_b \middle| \tilde{H}' \middle| \Psi_a \right\rangle dt \quad \text{(the 1st order solution, lecture 3)}$$

$$\Psi = \psi e^{-iE/\hbar} \qquad \omega_{ba} \equiv \frac{E_b - E_a}{\hbar}\Psi$$

$$c_b^{(1)}(T) = -\frac{i}{\hbar}\int_0^T \left\langle \psi_b \middle| \tilde{H}' \middle| \psi_a \right\rangle e^{i\omega_{ba}t}\, dt$$

20

Interaction with circularly polarized light

*Given the right combination of μ and **m**, absorption of R and L polarization is different → circular dichroism*

Recall lecture 3: tra

$$c_b^{(1)}(T) = -\frac{i}{\hbar}\int_0^T \langle \psi_b | \tilde{H}' | \psi_a \rangle e^{i\omega_{ba}t} dt \qquad \tilde{H}'(t) = -\boldsymbol{E}(t)\cdot\tilde{\mu} - \boldsymbol{B}(t)\cdot\tilde{m}$$

$$\langle \psi_b | \tilde{H} | \psi_a \rangle e^{i\omega_{ba}t} = \langle \psi_b | \tilde{E}\cdot\tilde{\mu} | \psi_a \rangle e^{i\omega_{ba}t} + \langle \psi_b | \tilde{B}\cdot\tilde{m} | \psi_a \rangle e^{i\omega_{ba}t}$$

$$= \tilde{E}\cdot\vec{\mu}_{ba} e^{i\omega_{ba}t} + \tilde{B}\cdot\vec{m}_{ba} e^{i\omega_{ba}t}$$

Circularly polarized light at z=0 (molecule is small): $\begin{cases} \tilde{E}^{\pm} = E_0(\hat{x} \pm i\hat{y})e^{-i\omega t} \\ \tilde{B}^{\pm} = \frac{1}{c}\hat{z}\times\tilde{E}^{\pm} = \mp\frac{i}{c}\tilde{E}^{\pm} \end{cases}$

+ = left
- = right

Assume resonance: $\omega=\omega_{ba}$

$$\langle \psi_b | \tilde{H} | \psi_a \rangle e^{i\omega_{ba}t} = E_0\left[(\hat{x} \pm i\hat{y})\cdot\vec{\mu}_{ba} \mp \frac{i}{c}(\hat{x} \pm i\hat{y})\cdot\vec{m}_{ba}\right]$$

$$|H_{ba}|^2 = E_0^2\left[(\hat{x} \mp i\hat{y})\cdot\vec{\mu}_{ba}^* \pm \frac{i}{c}(\hat{x} \mp i\hat{y})\cdot\vec{m}_{ba}^*\right]\left[(\hat{x} \pm i\hat{y})\cdot\vec{\mu}_{ba} \mp \frac{i}{c}(\hat{x} \pm i\hat{y})\cdot\vec{m}_{ba}\right]$$

21

Interaction with circularly polarized light

$$|H_{ba}|^2 = E_0^2\left[(\hat{x} \mp i\hat{y})\cdot\vec{\mu}_{ba}^* \pm \frac{i}{c}(\hat{x} \mp i\hat{y})\cdot\vec{m}_{ba}^*\right]\left[(\hat{x} \pm i\hat{y})\cdot\vec{\mu}_{ba} \mp \frac{i}{c}(\hat{x} \pm i\hat{y})\cdot\vec{m}_{ba}\right]$$

Notes:

$\vec{\mu} \equiv \vec{\mu}_{ba} = \vec{\mu}_{ba}^*$: the electric transition dipole is a real number (lectures 3-4)

$\vec{m} \equiv \vec{m}_{ba} = -\vec{m}_{ba}^*$: *m* is purely imaginary!

$$|H_{ba}|^2 = E_0^2\left[(\hat{x} \mp i\hat{y})\cdot\vec{\mu}_{ba} \mp \frac{i}{c}(\hat{x} \mp i\hat{y})\cdot\vec{m}_{ba}\right]\left[(\hat{x} \pm i\hat{y})\cdot\vec{\mu}_{ba} \mp \frac{i}{c}(\hat{x} \pm i\hat{y})\cdot\vec{m}_{ba}\right]$$

$$|H_{ba}|^2 = E_0^2\left[(\mu_x \mp i\mu_y) \mp \frac{i}{c}(m_x \mp im_y)\right]\left[(\mu_x \pm i\mu_y) \mp \frac{i}{c}(m_x \pm im_y)\right]$$

Four terms:

$$(\mu_x \mp i\mu_y)(\mu_x \pm i\mu_y) = \mu_x^2 + \mu_y^2 \qquad \mp\frac{i}{c}(\mu_x \mp i\mu_y)(m_x \pm im_y) = \mp\frac{i}{c}(\mu_x m_x \pm i\mu_x m_y \mp i\mu_y m_x + \mu_y m_y)$$

$$-\frac{1}{c^2}(m_x \mp im_y)(m_x \pm im_y) = \frac{1}{c^2}\left[m_x^2 + m_y^2\right] \qquad \mp\frac{i}{c}(m_x \mp im_y)\cdot(\mu_x \pm i\mu_y) = \mp\frac{i}{c}(\mu_x m_x \mp i\mu_x m_y \pm i\mu_y m_x + \mu_y m_y)$$

$$|\tilde{H}'_{ba}|^2 = E_0^2\left(\left[\mu_x^2 + \mu_y^2\right] \mp \frac{2i}{c}\left[\mu_x m_x + \mu_y m_y\right] + \frac{1}{c^2}\left[m_x^2 + m_y^2\right]\right)$$

22

Interaction with circularly polarized light

$$\left|\tilde{H}'_{ba}\right|^2 = E_0^2\left(\left[\mu_x^2+\mu_y^2\right] \mp \frac{2i}{c}\left[\mu_x m_x + \mu_y m_y\right] + \frac{1}{c^2}\left[m_x^2+m_y^2\right]\right)$$

$$\begin{cases} \tilde{E}^{\pm} = 2^{-1/2} E_0\left(\hat{x} \pm i\hat{y}\right)e^{-i\omega t} \\ \tilde{B}^{\pm} = \frac{1}{c}\hat{z}\times\vec{E}^{\pm} = \mp\frac{i}{c}\vec{E}^{\pm} \end{cases}$$

E and B are in the xy-plane

Assume transition moments are in the xy-plane (largest interaction):

$$\left|\tilde{H}'_{ba}\right|^2 = E_0^2\left|\vec{\mu}\right|^2 \mp 2iE_0B_0\vec{\mu}\cdot\vec{m} + B_0^2\left|\vec{m}\right|^2$$

For linear: $\left|\vec{E}\cdot\vec{\mu}\right|^2$, $\left|\vec{B}\cdot\vec{m}\right|^2$

$E_0^2\left|\vec{\mu}\right|^2$ Absorption due to the electric transition dipole moment, strength $D_E = \left|\vec{\mu}\right|^2$

$B_0^2\left|\vec{m}\right|^2$ Absorption due to the magnetic transition dipole moment, strength $D_B = \left|\vec{m}\right|^2$

$\mp 2iE_0B_0\vec{\mu}\cdot\vec{m}$ Absorption due to the magnetic and electric transition dipole moments $D_{L,R} = \mp 2\,\mathrm{Im}\left(\vec{\mu}\cdot\vec{m}\right)$

CD difference $\Delta A_{CD} \propto \left|\tilde{H}'_{ba}\right|_+^2 - \left|\tilde{H}'_{ba}\right|_-^2 = E_0B_0\left[-4\,\mathrm{Im}\left(\vec{\mu}_{ba}\cdot\vec{m}_{ba}\right)\right]$

Define Rotational Strength: $\boxed{\mathcal{R} \equiv -\mathrm{Im}\left(\vec{\mu}_{ba}\cdot\vec{m}_{ba}\right) = \frac{D_L - D_R}{4}}$

(Rosenfeld equation, 1928)
R is a real number

$$\boxed{D_L - D_R = 4\mathcal{R}}$$

The coefficient 4 stems from Rosenfeld's definition (apparently)

23

Interaction with circularly polarized light

		Maximum in optimal orientation	Averaging over all possible orientations: For circular	For linear	Strength
$\left|\vec{E}\cdot\vec{\mu}\right|^2$	$=$	$E_0^2 D_E$	$\frac{2}{3}E_0^2 D_E$	$\frac{1}{3}E_0^2 D_E$	$D_E \equiv \left|\vec{\mu}\right|^2$
$\left|\vec{B}\cdot\vec{m}\right|^2$	$=$	$B_0^2 D_B$	$\frac{2}{3}B_0^2 D_B$	$\frac{1}{3}B_0^2 D_B$	$D_B \equiv \left|\vec{m}\right|^2$
CD difference		$E_0B_0\left(4\mathcal{R}\right)$	$\frac{2}{3}E_0B_0\left(4\mathcal{R}\right)$		$\mathcal{R} = -\mathrm{Im}\left(\vec{\mu}_{ba}\cdot\vec{m}_{ba}\right)$

2/3 vs 1/3

Note: for the same E_0, the energy density of circularly polarized light is twice as large as for linearly polarized light, hence the 2x difference in averages for circular and linear polarizations..

If average absorption is expressed as a function of light energy density, i.e., use $1/\sqrt{2}$ *coefficient, the angular averaging coefficient is the same 1/3 for circular and linear polarizations!*

24

Rotational strength from experiment

Rotational strength: $\mathcal{R} = \dfrac{D_L - D_R}{4}$ (transition dipole strengths, quantum)

One can find D and R from experimental measurements of molar extinction spectra

D from absorption: $D = \dfrac{3hc \cdot 10^3 \ln(10)}{8\pi^3 N_A} \int \dfrac{\varepsilon(\tilde{\nu})}{\tilde{\nu}} d\tilde{\nu}$

R from absorption: $\mathcal{R} = \dfrac{3hc \cdot 10^3 \ln(10)}{32\pi^3 N_A} \int \dfrac{\Delta\varepsilon(\tilde{\nu})}{\tilde{\nu}} d\tilde{\nu}$

If you have a CD spectrum, you can estimate the rotational strength of a specific band using the *Gaussian approximation*:

$$\mathcal{R} \approx 1.23 \times 10^{-42} \cdot \frac{\Delta\varepsilon_{\max} \cdot \Delta\tilde{\nu}}{\tilde{\nu}_{\max}}$$

- $\Delta\varepsilon_{\max}$ is the peak molar circular dichroism.
- $\tilde{\nu}_{\max}$ is the wavenumber at the peak.
- $\Delta\tilde{\nu}$ is the half-width of the band at $1/e$ of its maximum height.

Also: $\dfrac{\mathcal{R}}{D} = \dfrac{\Delta A_{CD}}{4A} = -\dfrac{\mathrm{Im}\left(\vec{\mu}_{ba} \cdot \vec{m}_{ba}\right)}{\mu_{ba}^2} = -\dfrac{|m_{ba}|}{|\mu_{ba}|} \cos\alpha_{m\mu}$

25

Intrinsic CD: planar molecules

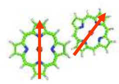
Planar molecules have no intrinsic CD
For flat (planar) molecules (such as chlorophyll), the magnetic transition dipole moment typically involves "circular" movement of electrons within the plane of the molecule (such as $\pi \rightarrow \pi^*$ transitions). Thus, the magnetic transition moment points perpendicular to the molecular plane, while the electric transition dipole is in the plane of the molecule → there is no intrinsic CD ($\mu \cdot \mathbf{m}$=0).

26

Oligomers: excitonic CD

Handwaving explanation

Example: a strongly coupled dimer



$H\Psi_i = E_i \Psi_i$

$H = \begin{pmatrix} E & V \\ V & E \end{pmatrix}$

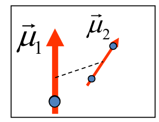
Delocalized states

$\Psi_1^* \sim |10\rangle + |01\rangle$

$\Psi_2^* \sim |10\rangle - |01\rangle$

$\vec{\eta}_1 \Rightarrow \vec{\mu}_1 + \vec{\mu}_2$

$\vec{\eta}_2 \Rightarrow \vec{\mu}_1 - \vec{\mu}_2$



The electric field direction is different at two locations

For in-phase transition: $\vec{\eta}_1 \Rightarrow \vec{\mu}_1 + \vec{\mu}_2$
Right circularly polarized light is absorbed better

For out-of-phase transition: $\vec{\eta}_2 \Rightarrow \vec{\mu}_1 - \vec{\mu}_2$
Left circularly polarized light is absorbed better

CD stems from the spatial distribution of coupled transition dipole moments
The main CD effect stems from electric moments; magnetic moments are weak

Oligomers: excitonic CD

$$c_b^{(1)}(T) = -\frac{i}{\hbar}\int_0^T \langle \Psi_b | \tilde{H}' | \Psi_a \rangle dt$$

Define: $\tilde{H}'_{ba} \equiv \langle \Psi_b | \tilde{H}' | \Psi_a \rangle$

Dipole operator

Ground state $\varphi_a(\vec{r}_i)e^{-i\omega_a t}$

Interaction with light: $\tilde{H}'_{ba} = \vec{E}(\vec{r},t)\cdot\langle \Psi_b | \hat{\mu} | \Psi_a \rangle$ (Lecture 4)

Excited state

Stationary excited excitonic (delocalized) state n:

Exciton state k energy $\omega_n = E_n / \hbar$

$$\Psi_n(\vec{r},t) = \sum_i c_{ni}\varphi_i(\vec{r}_i)e^{-i\omega_n t}$$ (Lecture 11)

Excited states of individual molecules (basis)

Circularly polarized light propagating along the z-axis: $\vec{E}^{\pm} = \frac{1}{\sqrt{2}} E_0 (\hat{x} \pm i\hat{y}) e^{i(kz-\omega t)}$

+ left, - right

Then for the excited state *kn*:

$$\tilde{H}'_{na} = \sum_i c_{ni}\frac{E_0}{\sqrt{2}}(\hat{x} \pm i\hat{y})e^{i(kz_i-\omega t)}\cdot\langle \varphi_i^*(\vec{r}_i)e^{i\omega_n t} | \hat{\mu} | \varphi_a(\vec{r}_i)e^{-i\omega_a t}\rangle$$

z-component of the chromophore's position

Oligomers: excitonic CD

$$\tilde{H}'_{na} = 2^{-1/2} E_0 \sum_i c_{ni} \left(\hat{x} \pm i\hat{y}\right) e^{i(kz_i - \omega t)} \cdot \left\langle \varphi_i^* \left(\vec{r}_i\right) e^{i\omega_n t} \left|\hat{\mu}\right| \varphi_a \left(\vec{r}_i\right) e^{-i\omega_a t} \right\rangle$$

Simplify:

$\omega_n - \omega_a = \omega_{na} = E_{na} / \hbar$ Transition energy

$$\tilde{H}'_{na} = 2^{-1/2} E_0 \sum_i c_{ni} \left(\hat{x} \pm i\hat{y}\right) e^{i(kz_i - \omega t)} \cdot e^{i(\omega_n - \omega_a)t} \left\langle \varphi_i^* \left(\vec{r}_i\right) \left|\hat{\mu}\right| \varphi_a \left(\vec{r}_i\right) \right\rangle$$

Transition dipole moment of chromophore i (lecture 4)

$$= 2^{-1/2} E_0 \sum_i c_{ni} \left(\hat{x} \pm i\hat{y}\right) e^{ikz_i} \cdot \vec{\mu}_i e^{i(\omega_{na} - \omega)t}$$

Assume resonance excitation: $\omega_{na} = \omega$ (largest integral over t)

$$= 2^{-1/2} E_0 \sum_i c_{ni} \left(\hat{x} \pm i\hat{y}\right) e^{ikz_i} \cdot \vec{\mu}_i$$

$$c_n^{(1)}\left(T\right) = -\frac{i}{\hbar} \int_0^T \tilde{H}'_{na} dt$$

Transition strength is proportional to: $\left|\tilde{H}'_{na}\right|^2 = \tilde{H}'^*_{na} \tilde{H}'_{na}$

$$\left|\tilde{H}'_{na}\right|^2 = \tfrac{1}{2} \left|E_0\right|^2 \left[\sum_i c_{ni}^* \left(\hat{x} \mp i\hat{y}\right) \cdot \vec{\mu}_i e^{-ikz_i} \right] \left[\sum_j c_{nj} \left(\hat{x} \pm i\hat{y}\right) \cdot \vec{\mu}_j e^{ikz_j} \right]$$

$$= \tfrac{1}{2} \left|E_0\right|^2 \sum_i \sum_j c_{ni}^* c_{nj} \left[\left(\hat{x} \mp i\hat{y}\right) \cdot \vec{\mu}_i \right] \left[\left(\hat{x} \pm i\hat{y}\right) \cdot \vec{\mu}_j \right] e^{ik\left(z_j - z_i\right)}$$

29

Oligomers: excitonic CD

$$\left|\tilde{H}'_{na}\right|^2 = \tfrac{1}{2} \left|E_0\right|^2 \sum_i \sum_j c_{ni}^* c_{nj} \left[\left(\hat{x} \mp i\hat{y}\right) \cdot \vec{\mu}_i \right] \left[\left(\hat{x} \pm i\hat{y}\right) \cdot \vec{\mu}_j \right] e^{-ikz_{ij}}$$

$z_{ij} \equiv z_i - z_j$ *from j to i*

$$\left|\tilde{H}'_{na}\right|^2 = \tfrac{1}{2} \left|E_0\right|^2 \sum_i \sum_j c_{ni}^* c_{nj} \left[\left(\mu_{ix} \mp i\mu_{iy}\right) \left(\vec{\mu}_{jx} \pm i\mu_{jy}\right) \right] e^{-ikz_{ij}}$$

$$= \tfrac{1}{2} \left|E_0\right|^2 \sum_i \sum_k c_{ni}^* c_{nj} \left[\mu_{ix}\mu_{jx} \pm i\mu_{ix}\mu_{jy} \mp i\mu_{iy}\mu_{jx} + \mu_{iy}\mu_{jy} \right] e^{-ikz_{ij}}$$

Cancel in difference

Circular dichroism: $\Delta A_{CD} = A_L - A_R \propto \Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|\tilde{H}'_{na}\right|_+^2 - \left|\tilde{H}'_{na}\right|_-^2$

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_k c_{ni}^* c_{nj} \left[i\left(\vec{\mu}_{ix}\vec{\mu}_{jy} - \vec{\mu}_{iy}\vec{\mu}_{jx}\right) \right] e^{-ikz_{ij}}$$

z-component of cross-product $\vec{\mu}_i \times \vec{\mu}_j$

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_j c_{ni}^* c_{nj} \left[i\hat{z} \cdot \left(\vec{\mu}_i \times \vec{\mu}_j\right) \right] e^{-ikz_{ij}}$$

30

Oligomers: excitonic CD

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[i\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] e^{-ikz_{ij}} \qquad z_{ij} \equiv z_i - z_j$$

$$e^{-ikz_{ij}} = \cos\left(kz_{ij}\right) - i\sin\left(kz_{ij}\right)$$

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[i\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] \cos kz_{ij} + \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] \sin kz_{ij}$$

Notes: $\left(\vec{\mu}_i \times \vec{\mu}_j\right) = -\left(\vec{\mu}_j \times \vec{\mu}_i\right)$ and $z_{ji} \equiv -z_{ij}$

Sums always have terms with ij and ji;
The first sum is odd with respect to ij to ji substitution, → it will be 0:

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] \sin kz_{ij}$$

The distance between interacting pigments is short, z_{ji} ~10 Å, while the wavelength of light is λ~6000 Å → kz_{ji} ~ 1/100. For small angles sin(x)~x:

$$k = \frac{2\pi}{\lambda}$$

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} \approx \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] kz_{ij}$$

31

Oligomers: excitonic CD

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] kz_{ij} \qquad z_{ij} \equiv z_i - z_j = \hat{z}\cdot\vec{r}_{ij}$$

Direction from j to i

$$\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD} = \frac{2\pi}{\lambda}\left|E_0\right|^2 \sum_i \sum_j c^*_{ni} c_{nj} \left[\hat{z}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right] \hat{z}\cdot\vec{r}_{ij} \qquad k = \frac{2\pi}{\lambda}$$

Average over all possible directions of z: $\overline{\left(\hat{z}\cdot\vec{A}\right)\left(\hat{z}\cdot\vec{B}\right)} = \frac{1}{3}\vec{A}\cdot\vec{B}$

$$\overline{\Delta\left(\left|\tilde{H}'_{na}\right|^2\right)_{CD}} = \left|E_0\right|^2 \frac{2\pi}{3\lambda} \sum_i \sum_j c^*_{ni} c_{nj} \vec{r}_{ij}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)$$

This is the difference in apparent transition dipole strength for left (D_L) and right (D_R) circularly polarized light absorption averaged over all directions. The rotational strength is defined as

$$\mathcal{R} = \frac{D_L - D_R}{4}$$

And the averaging coefficient (1/3) is dropped (as earlier for intrinsic CD):

Rotational strength

$$\boxed{\mathcal{R}_n = \frac{\pi}{2\lambda} \sum_{i,j} c_{ni} c_{nj} \left[\vec{r}_{ij}\cdot\left(\vec{\mu}_i \times \vec{\mu}_j\right)\right]}$$

32

Oligomers: excitonic CD

Rotational strength for excitonic transition *n*:

$$\mathcal{R}_n = \frac{\pi}{2\lambda}\sum_{i,j} c_{ni}c_{nj}\left[\vec{r}_{ij}\cdot\left(\vec{\mu}_i\times\vec{\mu}_j\right)\right]$$

$\vec{r}_{ij} \equiv \vec{r}_i - \vec{r}_j$

Vector from j to i

if i=j, r_{ij} = 0

Transition dipole strength for excitonic transition:

$$D_n = \left|\sum_i c_{ni}\vec{\mu}_i\right|^2 = \sum_{i,j} c_{ni}c_{nj}\vec{\mu}_i\cdot\vec{\mu}_j$$

To average over all possible directions x1/3

Notes on oligomeric CD:

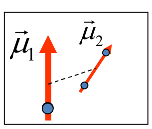
- Must superimpose the effects of intrinsic molecular $\mathcal{R}$ and excitonic $\mathcal{R}$
- For a flat molecule (like chlorophyll), there is no intrinsic CD
- We ignored CD due to couplings between magnetic dipole moments and magnetic-electric dipole moments for different molecules (they are typically negligible compared to excitonic CD)
- *The actual ΔA_{CD} for a single oligomer oriented in a specific way could be positive or negative, even if the (averaged) $\mathcal{R}$ is zero!*
- The excitonic CD spectrum is conservative, i.e., integrated intensity is zero

33

Excitonic CD: homodimer example

$$\mathcal{R} = \frac{\pi}{2\lambda}\sum_{i,j} c_{ki}c_{kj}\left[\vec{r}_{ij}\cdot\left(\vec{\mu}_i\times\vec{\mu}_j\right)\right]$$

Delocalized states



$$\Psi_+ = \frac{1}{\sqrt{2}}\psi_1 + \frac{1}{\sqrt{2}}\psi_1 \qquad \vec{\mu}_+ = \frac{1}{\sqrt{2}}\vec{\mu}_1 + \frac{1}{\sqrt{2}}\vec{\mu}_2$$

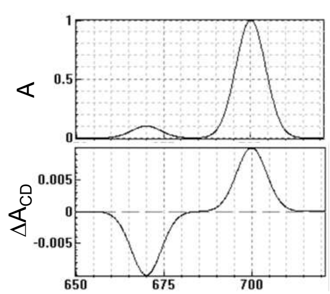
$$\Psi_- = \frac{1}{\sqrt{2}}\psi_1 - \frac{1}{\sqrt{2}}\psi_1 \qquad \vec{\mu}_- = \frac{1}{\sqrt{2}}\vec{\mu}_1 - \frac{1}{\sqrt{2}}\vec{\mu}_2$$

Note: if i=j, r_{ij} = 0

$$\mathcal{R}_\pm = \frac{\pi}{2\lambda_\pm}\left[\pm\frac{1}{2}\vec{r}_{12}\cdot\left(\vec{\mu}_1\times\vec{\mu}_2\right)\pm\frac{1}{2}\vec{r}_{21}\cdot\left(\vec{\mu}_2\times\vec{\mu}_1\right)\right]$$

$$\vec{r}_{21} = -\vec{r}_{12} \qquad \vec{\mu}_2\times\vec{\mu}_1 = -\vec{\mu}_1\times\vec{\mu}_2$$

$$\mathcal{R}_\pm = \pm\frac{\pi}{2\lambda_\pm}\vec{r}_{12}\cdot\left(\vec{\mu}_1\times\vec{\mu}_2\right) \qquad \vec{r}_{12} \equiv \vec{r}_1 - \vec{r}_2$$

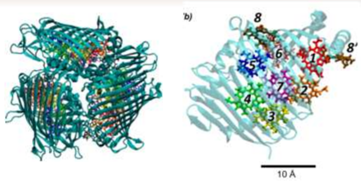


While the absorption strengths of each band could be different, the CD signals have the same magnitudes but opposite signs

→ CD spectra can distinguish both bands even if one is 'dark.'
→ Since bands are positive and negative, one can distinguish overlapping bands

34

Excitonic CD: the Fenna Matthews Olson complex



1. Hamiltonian in the basis of pigment's wavefunctions

$$H=\begin{bmatrix} 12420 & -106 & 8 & -5 & 6 & -8 & -4 \\ -106 & 12560 & 28 & 6 & 2 & 13 & 1 \\ 8 & 28 & 12140 & -62 & -1 & -9 & 17 \\ -5 & 6 & -62 & 12335 & -70 & -19 & -57 \\ 6 & 2 & -1 & -70 & 12460 & 40 & -2 \\ -8 & 13 & -9 & -19 & 40 & 12500 & 32 \\ -4 & 1 & 17 & -57 & -2 & 32 & 12400 \end{bmatrix} \quad \boldsymbol{\psi}_n = \begin{bmatrix} c_{n1} \\ c_{n2} \\ \vdots \\ c_{n7} \end{bmatrix}$$

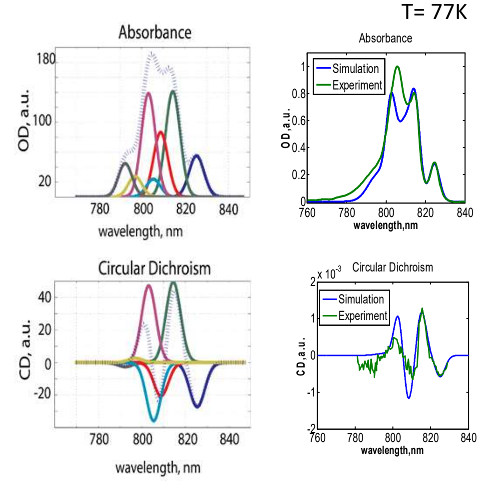
2. Solve TISE: $\mathbf{H}\boldsymbol{\psi}_n = E_n \boldsymbol{\psi}$

Obtain (by **H** diagonalization):
E_n – exciton transition energies (eigenvalues)
c_{ni} – expansion coefficients (eigenvectors)

3. Calculate: $D_e = \left| \sum_{n,m=1}^{N} c_{en}\vec{\mu}_n \right|^2 = \sum_{n,m=1}^{N} c_{en}c_{em}^{*}\vec{\mu}_n\vec{\mu}_m$

$$R_e = \sum_{n,m=1}^{N} c_{en}c_{em}^{*}\vec{r}_{nm} \bullet (\vec{\mu}_n \times \vec{\mu}_m)$$

T= 77K



35

Making circularly polarized light

How to make circularly polarized light?

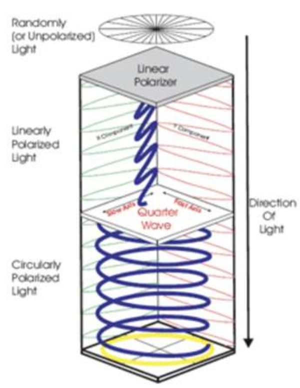
$$k = \frac{2\pi}{\lambda}$$

$$\vec{E}^{\pm}(t,z) = E_0\left[\hat{x}\cos(\omega t - kz) \pm \hat{y}\sin(\omega t - kz)\right]$$
$$= E_0\left[\hat{x}\cos(\omega t - kz) + \hat{y}\cos(\omega t - kz + \theta)\right]$$

$\theta=\pm\pi/2$ - Phase shift for one axis results in circular polarization (shift of π will rotate polarization by 90 degrees)
$\theta=\pm\pi/2$ corresponds to path delay by quarter wavelength, $\lambda/4$

$$\vec{E}^{\pm}(t,z) = E_0\left[\hat{x}\cos(\omega t - kz) + \hat{y}\cos(\omega t - k[z + n_\theta\lambda])\right]$$

n_θ - delay in wavelengths



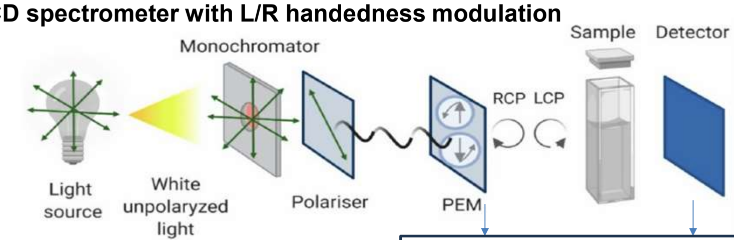
Wave-plate: refraction index is different for light polarized along orthogonal axes
Quarter-wave plate: difference in optical path is $\lambda/4$ (+$n\lambda$), n_θ=1/4. By rotating $\lambda/4$-plate by 90 degrees, one can switch between L and R polarizations

Half-wave plate: can rotate polarization by any angle up to 90 degrees

36

Measuring circular dichroism

CD spectrometer with L/R handedness modulation



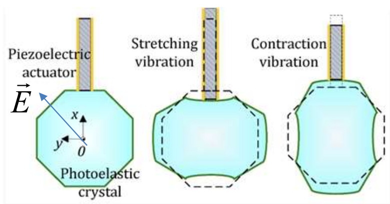
Reference $\sin(\omega t)$ Signal $S(t)$

Lock-in amplifier

Detects the signal at a certain frequency

Fourier trick: $\Delta T_{CD} = a_\omega = \int S(t)\sin(\omega t)\,dt$

PEM, Photo-Elastic Modulator: Soundwave modulates the density of glass and the refractive index



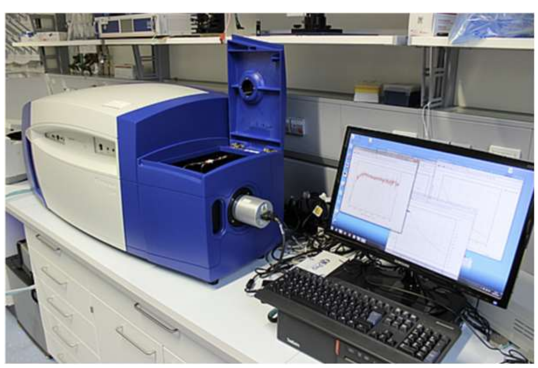
Compressing glass along one axis changes the refractive index (and speed of light)

Typically, the standing acoustic wave frequency is ~50 kHz, so light alternates between R and L at this frequency.

37

CD spectrometer

Push-button, no brains needed to operate



ZEMOS CD spectrometer

38

EXTRA

39

Interaction with circularly polarized light

Generated by Gemini AI

$$\widetilde{\mathrm{H}}'(t) = -\boldsymbol{E}(t)\cdot\widetilde{\mu} - \boldsymbol{B}(t)\cdot\widetilde{\mathrm{m}}$$

OVERSIMPLIFIED!!!

Transition probability is: $W_{ba} \propto \left|\tilde{H}'_{ba}\right|^2$

Note: assume that the light frequency is in resonance with the transition

$$\tilde{H}'_{ba} = \left\langle \psi_b \middle| -\vec{E}\cdot\tilde{\mu} - \vec{B}\cdot\tilde{m} \middle| \psi_a \right\rangle = -\vec{E}\left\langle \psi_b \middle| \tilde{\mu} \middle| \psi_a \right\rangle - \vec{B}\left\langle \psi_b \middle| \tilde{m} \middle| \psi_a \right\rangle = -\vec{E}\cdot\vec{\mu}_{ba} - \vec{B}\cdot\vec{m}_{ba}$$

Circularly polarized light: $\begin{cases} \vec{E}^{\pm} = E_0\left(\hat{x} \pm i\hat{y}\right)e^{i(kz-\omega t)} \\ \vec{B}^{\pm} = \dfrac{1}{c}\hat{z}\times\vec{E}^{\pm} = \mp\dfrac{i}{c}\vec{E}^{\pm} \end{cases}$

+ = left
- = right

Circular dichroism: $\Delta A_{CD} = W_{ba}^{+} - W_{ba}^{-}$

$$\left|\tilde{H}'^{\pm}_{ba}\right|^2 = \left|-\vec{E}^{\pm}\cdot\vec{\mu}_{ba} \pm \frac{i}{c}\vec{E}^{\pm}\cdot\vec{m}_{ba}\right|^2$$

WRONG!

$$\left|\tilde{H}'^{\pm}_{ba}\right|^2 = E_0^2\left(\left|\vec{\mu}_{ba}\right|^2 \pm \left|\frac{2i}{c}\vec{\mu}_{ba}\cdot\vec{m}_{ba}\right| + \frac{1}{c^2}\left|\vec{m}_{ba}\right|^2\right)$$

The first and last terms are the same for the right and left polarizations:

$$R = \mathrm{Im}\left(\vec{\mu}_{ba}\cdot\vec{m}_{ba}\right)$$

BUT RIGHT ANSWER

40

Gemini: rotational strength in radians vs absorption difference of left and right polarized light, equation

- **$\Delta\epsilon$**: $\epsilon_L - \epsilon_R$ (Molar circular dichroism, L/mol·cm).
- **λ**: Wavelength (nm).

Key Relationships

- **Absorption Difference (ΔA):** The direct experimental measurement is $\Delta A = A_L - A_R$.
- **Ellipticity (θ) and Radians:** If measured as ellipticity θ in radians, it is related to absorbance difference by $\tan\theta = \frac{E_R - E_L}{E_R + E_L} \approx \Delta A \frac{\ln(10)}{4}$.
- **Small Angle Approximation:** Since ΔA is small, the ellipticity in radians is often approximated directly from the absorbance difference:

$$\Delta A \approx \frac{4\theta_{\text{rad}}}{\ln(10)}$$

Wikipedia +1

Summary Table of Units

Quantity	Common Unit	Relationship to R
Rotational Strength (R)	10^{-40} cgs ($Debye \cdot BM$)	Fundamental parameter
CD ($\Delta\epsilon$)	$L \cdot mol^{-1} \cdot cm^{-1}$	$\int \Delta\epsilon dv$
Ellipticity (θ)	Millidegrees (m°) or Radians	$\Delta A \cdot 32.982 \approx m^\circ$

Key Takeaway: The rotational strength is the integrated value of the difference in absorption ($\epsilon_L - \epsilon_R$) scaled by a factor that includes fundamental physical constants, allowing us to relate light-matter interaction to molecular geometry. Northern Illinois University

41

1. The "1/4" Rule is for Dipole Strengths (D)

The factor of $1/4$ applies when you are looking at the **Dipole Strength** (the transition's intensity), which has units of D^2 (Debye squared):

$$R = \frac{D_L - D_R}{4}$$

Here, D_L and D_R are the dipole strengths for left and right polarized light, respectively.

2. The Relationship to Molar Circular Dichroism ($\Delta\varepsilon$)

Rotational strength R is related to the difference in molar absorption ($\Delta\varepsilon = \varepsilon_L - \varepsilon_R$), but it requires an **integral over the entire absorption band** because R represents the total "area" of the transition, not a single point value.

The standard conversion is:

$$R \propto \int \frac{\varepsilon_L(\nu) - \varepsilon_R(\nu)}{\nu}\, d\nu$$

42

1. Fundamental Integral Relationship

The most common form of this relationship uses **molar circular dichroism** ($\Delta\varepsilon = \varepsilon_L - \varepsilon_R$) and the wavenumber ($\tilde{\nu}$):

$$R = \frac{3hc \cdot 10^3 \ln(10)}{32\pi^3 N_A} \int \frac{\Delta\varepsilon(\tilde{\nu})}{\tilde{\nu}} \, d\tilde{\nu}$$

In standard cgs units (where R is in 10^{-40} erg·cm^3 or Debye-Bohr magnetons), this simplifies to:

$$R \approx 2.296 \times 10^{-39} \int \frac{\Delta\varepsilon(\tilde{\nu})}{\tilde{\nu}} \, d\tilde{\nu}$$

2. Gaussian Approximation for a Single Band

If the absorption difference band follows a Gaussian distribution, you can convert the rotational strength directly from the peak properties of the band without a complex integral:

$$R \approx 1.23 \times 10^{-42} \cdot \frac{\Delta\varepsilon_{\max} \cdot \Delta\tilde{\nu}}{\tilde{\nu}_{\max}}$$

Summary of Units

- $\boldsymbol{R}$: Rotational Strength (units of 10^{-40} erg·cm^3 or 1 Debye-Bohr magneton $\approx 0.9273 \times 10^{-38}$ erg·cm^3).
- $\boldsymbol{\Delta\varepsilon}$: Molar circular dichroism ($M^{-1}\mathrm{cm}^{-1}$).
- $\boldsymbol{\tilde{\nu}}$: Wavenumber (cm^{-1}).

43

The relationship between the **dipole strength** (D) and the **molar absorption coefficient** (ε) is an integral relationship, just like the one for rotational strength.

In simple terms, D is the **total area** under the absorption curve, while ε is the **height** of the curve at a specific frequency.

1. The Fundamental Equation

The dipole strength of an electronic transition is proportional to the integral of the molar absorption coefficient (ε) over the absorption band:

$$D = \frac{3hc \cdot 10^3 \ln(10)}{8\pi^3 N_A} \int \frac{\varepsilon(\tilde{\nu})}{\tilde{\nu}} \, d\tilde{\nu}$$

In standard units (where D is in **Debye^2** and $\tilde{\nu}$ is in **cm^{-1}**), this simplifies to the commonly used numerical constant:

$$D \approx 9.184 \times 10^{-3} \int \frac{\varepsilon(\tilde{\nu})}{\tilde{\nu}} \, d\tilde{\nu}$$

2. Gaussian Approximation

If the absorption peak looks like a standard bell curve (Gaussian), you can estimate D without doing a complex integral:

$$D \approx 9.184 \times 10^{-3} \, \frac{\varepsilon_{\max} \cdot \Delta\tilde{\nu}_{1/2}}{\tilde{\nu}_{\max}}$$

44

Directional Properties

- **Orbital Magnetic Moment**: For a pure p_z orbital ($m_l = 0$), the orbital magnetic moment along the z-axis is **zero**. This is because the electron has no net angular momentum about the z-axis in this state.
- **Transition Magnetic Dipole Moment**: If a transition occurs (e.g., from a p_x to a p_y orbital), the transition magnetic moment vector is typically directed **perpendicular** to the plane containing the transition's orbital movement.
- **Transition Electric Dipole Moment**: For a common electronic transition like $s \to p_z$, the transition dipole moment is directed strictly along the **z-axis.** Reddit +5

45

In a flat (planar) molecule, the **transition magnetic moment** ($\vec{m}$) **typically points perpendicular to the molecular plane**. This directionality is tied to the way charge "rotates" during a transition, similar to how current in a flat loop of wire generates a magnetic field normal to the loop's surface. Physics LibreTexts +1

Directional Characteristics

- **Perpendicular Orientation**: For transitions that involve "circular" movement of electrons within the plane (such as $\pi \to \pi^*$ or certain $n \to \pi^*$ transitions), the resulting magnetic moment is directed along the axis normal to the molecular plane.
- **Magnetic vs. Electric Dipole**: While the **transition electric dipole moment** ($\vec{\mu}$) usually lies **in-plane** for $\pi \to \pi^*$ transitions, the magnetic counterpart $\vec{m}$ is out-of-plane.
- **Chirality and Optical Activity**: In achiral flat molecules like formaldehyde or benzene, the direction of $\vec{m}$ is often zero or arbitrarily defined unless symmetry is broken. In chiral systems or supramolecular stacks, a strong out-of-plane $\vec{m}$ is a key indicator of **circular dichroism** and magnetic-dipole-allowedness. National Institutes of Health (.gov) +4

46

Key Physical Principles

1. **Right-Hand Rule**: The direction of $\vec{m}$ is determined by the "rotation" of charge density during the transition; using your right hand, fingers curling with the charge rotation will point the thumb in the direction of the magnetic moment.
2. **Symmetry Constraints**: In many planar molecules, transitions that are **electrically forbidden** (where $\vec{\mu} = 0$) can be **magnetically allowed** (where $\vec{m} \neq 0$), specifically if the transition involves a change in orbital angular momentum that mimics a current loop.
3. **Magnitude**: Typical values for strong magnetic transitions in flat aromatic systems are around **0.3 to 0.4 Bohr magnetons** (μ_B). National Institutes of Health (.gov) +4

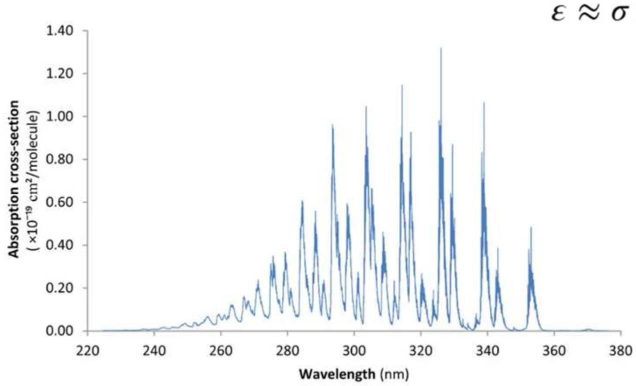
47

Formaldehyde (CH_2O):

In its n→π^* transition, the transition is strictly electric-dipole forbidden by symmetry. It becomes observable primarily through magnetic dipole transitions or by "stealing" intensity via molecular vibrations (vibronic coupling).

Convert to extinction:

$$\varepsilon \approx \sigma \times (2.615 \times 10^{20})$$



Formaldehyde absorption spectrum in the UV range, re-plotted from the data of Meller et al [30]. Peaks are not fully resolved at the resolution of this spectrum (0.01 nm).

48